
Flujo de inferencia

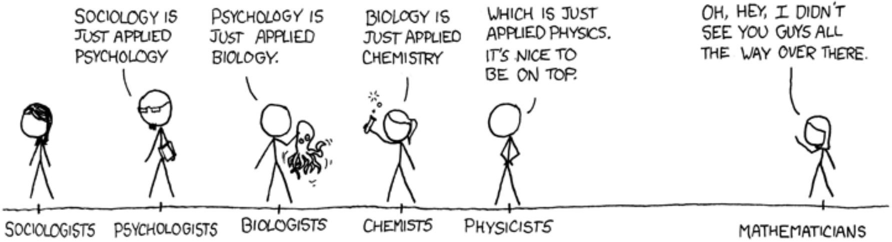
Bibliografía Unidad 4

- Bishop, C. **Pattern Recognition and Machine Learning**. 2006. (Descargar). (lectura 8.1, 8.2 y 8.4)
- Neal. **Pattern Recognition and Machine Learning**. 2020 (Draft). (Descargar). (lectura capítulo 1 y 3)

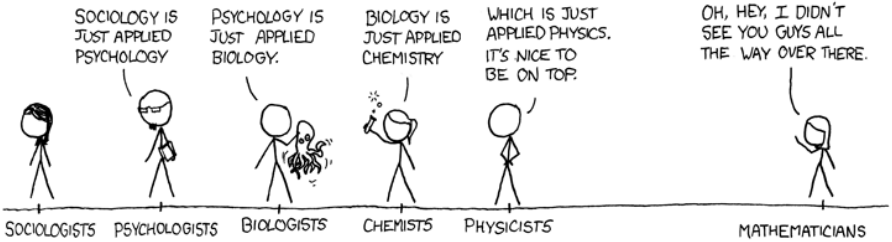
Otros:

- Kschischang. *Factor graphs and the sum-product algorithm*; IEEE Transactions on information theory. 2001. (Descargar). (lectura partes del paper)

Ciencias con datos Ciencias formales



Ciencias con datos Ciencias formales



Todas las ciencias con datos desarrollan **teorías causales** para comprender el mundo

La ventaja de las teorías causales

La ventaja de las teorías causales

$$-\overbrace{\log \lim_{T\to\infty} P(\textsf{Datos}_T|\textsf{Modelo Causal})^{1/T}}^{\substack{\textsf{Tasa de predicción en}\\ \textsf{órdenes de magnitud}}}$$

La ventaja de las teorías causales

$$\overbrace{-\log \lim_{T\to\infty} P(\text{Datos}_T|\text{Modelo Causal})^{1/T}}^{\text{Tasa de predicción en órdenes de magnitud}}$$

$$=$$

$$\underbrace{\sum_{c,s,r} \underbrace{P(c,s,r|\text{Realidad Causal})}_{\text{Probabilidad de que se genere el dato}} \cdot \underbrace{\big(-\log P(c,s,r|\text{Modelo Causal})\big)}_{\text{Información en órdenes de magnitud}}}_{\text{Entropía cruzada}}$$

La ventaja de las teorías causales

$$\overbrace{-\log \lim_{T\to\infty} P(\text{Datos}_T|\text{Modelo Causal})^{1/T}}^{\substack{\text{Tasa de predicción en}\\ \text{órdenes de magnitud}}}$$

$$=$$

$$\underbrace{\sum_{c,s,r} \underbrace{P(c,s,r|\text{Realidad Causal})}_{\substack{\text{Probabilidad de que}\\ \text{se genere el dato}}} \cdot \underbrace{\big(-\log P(c,s,r|\text{Modelo Causal})\big)}_{\substack{\text{Información en}\\ \text{órdenes de magnitud}}}}_{\text{Entropía cruzada}}$$

No hay mejor predicción que la del modelo causal

que se corresponde con la realidad causal subyacente

Predecir el impacto de las acciones

a partir de datos observados

$P(Y|X)$

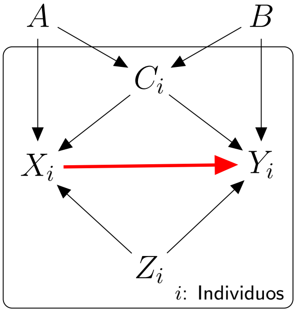
Predecir el impacto de las acciones

a partir de datos observados

$$P(Y|X) \qquad \neq \qquad P(Y|\mathsf{do}(X))$$

Predecir el impacto de las acciones
a partir de datos observados

Observaciones
$P(Y|X)$ $\neq$ $P(Y|\text{do}(X))$

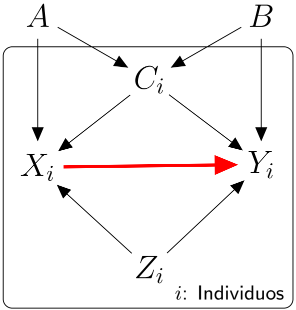


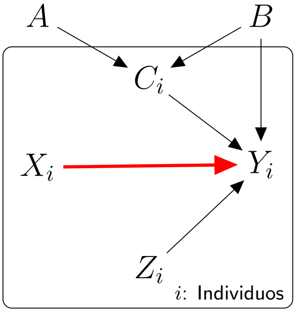
Predecir el impacto de las acciones
a partir de datos observados

Observaciones
$P(Y|X)$

$\neq$

Intervenciones
$P(Y|\text{do}(X))$





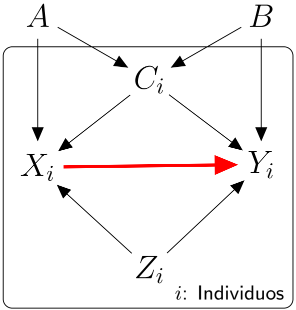
Predecir el impacto de las acciones

a partir de datos observados

Entrenamos

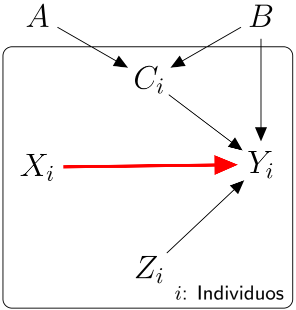
In-sample

Observaciones
$P(Y|X)$



$\neq$

Intervenciones
$P(Y|\text{do}(X))$



Predecir el impacto de las acciones

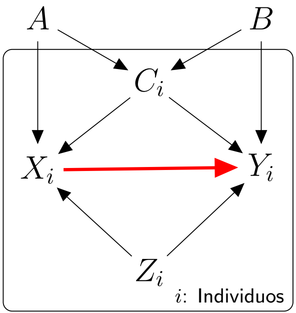
a partir de datos observados

Entrenamos

In-sample

Observaciones

$P(Y|X)$



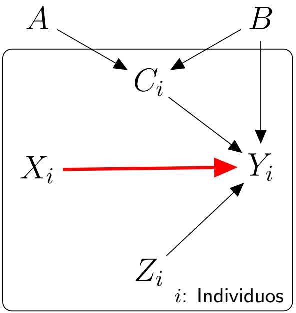
$\neq$

Predecimos

Out-of-sample

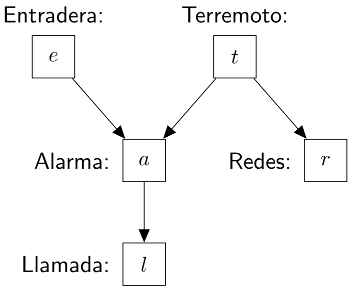
Intervenciones

$P(Y|\text{do}(X))$



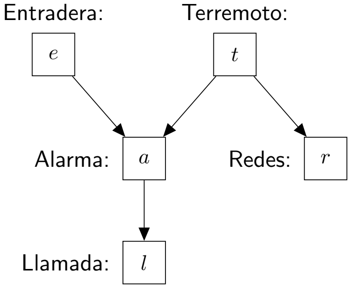
Flujo de inferencia

Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \overset{?}{=} P(r\|a)$	$P(r\|t) \overset{?}{=} P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \overset{?}{=} P(l\|e)$	$P(l\|a) \overset{?}{=} P(l\|e,a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \overset{?}{=} P(t\|e)$	$P(t\|a) \overset{?}{=} P(t\|e,a)$ $P(t\|l) \overset{?}{=} P(t\|e,l)$

Flujo de inferencia
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \overset{?}{=} P(r\|a)$	$P(r\|t) \overset{?}{=} P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \overset{?}{=} P(l\|e)$	$P(l\|a) \overset{?}{=} P(l\|e,a)$
Collider:	$\begin{matrix} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{matrix}$	$P(t) \overset{?}{=} P(t\|e)$	$P(t\|a) \overset{?}{=} P(t\|e,a)$ $P(t\|l) \overset{?}{=} P(t\|e,l)$

Flujo de inferencia

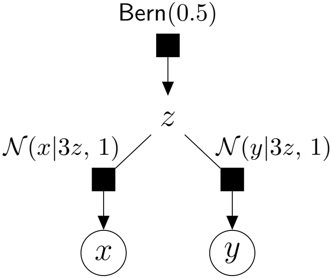
Fork

$$x \longleftarrow z \longrightarrow y$$

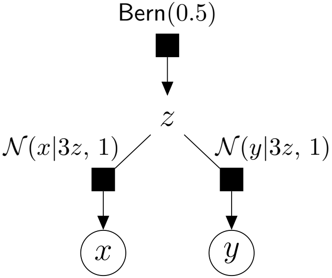
Flujo de inferencia

Fork

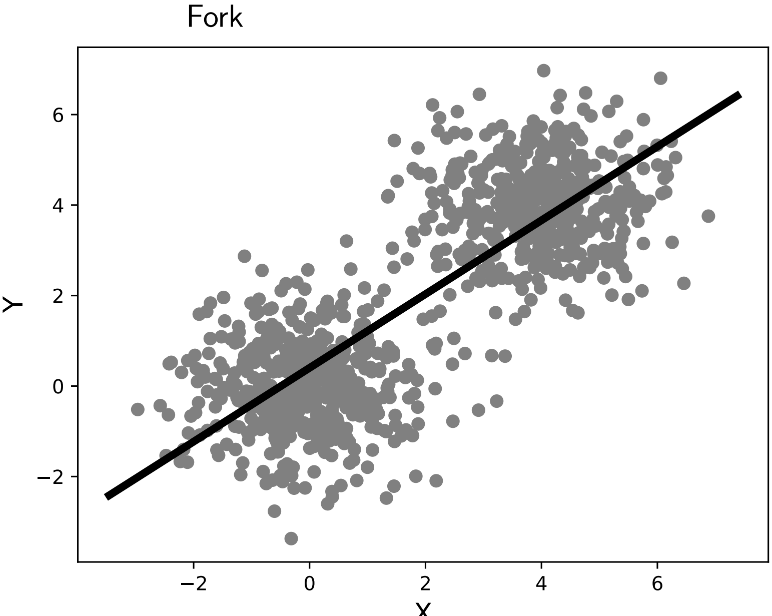
$$x \longleftarrow z \longrightarrow y$$



Flujo de inferencia

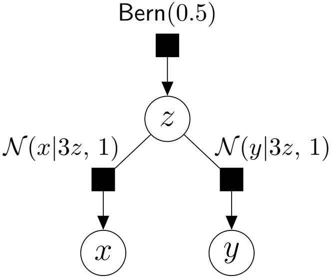


$$X \not\perp\!\!\!\perp Y | \emptyset$$

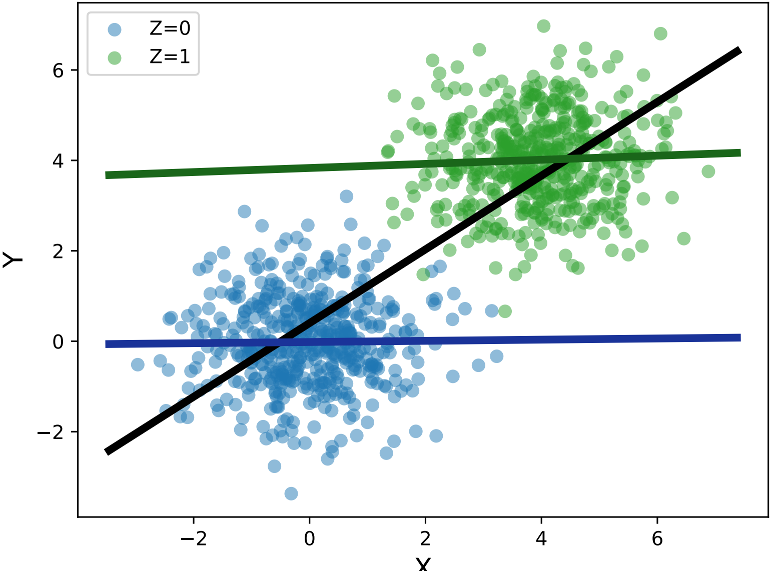


Flujo de inferencia

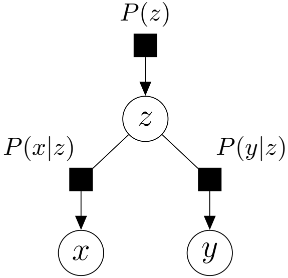
Fork



$$X \perp\!\!\!\perp Y | Z$$



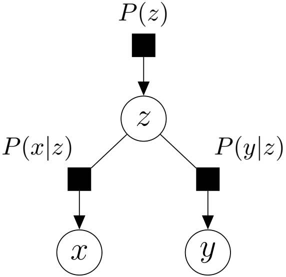
Flujo de inferencia
Fork



$$X \perp\!\!\!\perp Y|Z \iff P(x,y|z) = P(x|z)P(y|z)$$

Flujo de inferencia

Fork

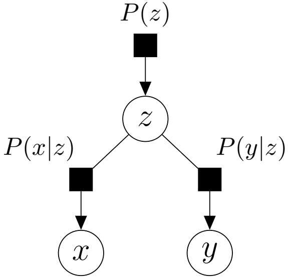


$$X \perp\!\!\!\perp Y | Z$$

$$P(x, y|z) =$$

Flujo de inferencia

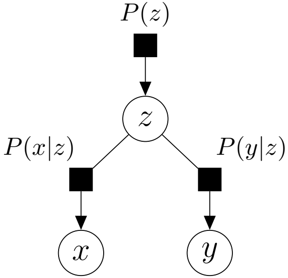
Fork



$$X \perp\!\!\!\perp Y | Z$$

$$P(x, y|z) = \frac{P(x, y, z)}{P(z)}$$

Flujo de inferencia
Fork

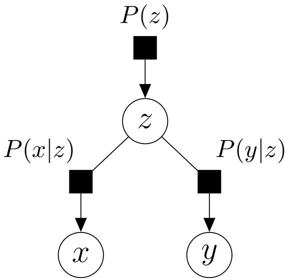


$$X \perp\!\!\!\perp Y|Z$$

$$P(x,y|z) = \frac{P(x|z)P(y|z)P(z)}{P(z)}$$

Flujo de inferencia

Fork

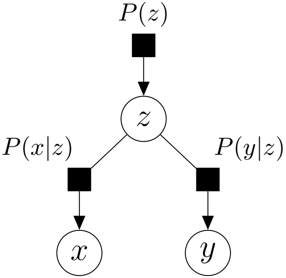


$$X \perp\!\!\!\perp Y | Z$$

$$P(x,y|z) = \frac{P(x|z)P(y|z)\cancel{P(z)}}{\cancel{P(z)}}$$

Flujo de inferencia

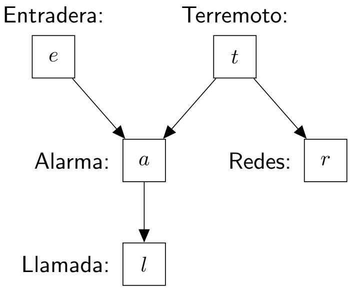
Fork



$$X \perp\!\!\!\perp Y | Z$$

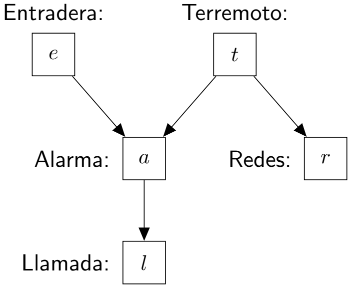
$$P(x, y|z) = \frac{P(x|z)P(y|z)\cancel{P(z)}}{\cancel{P(z)}} = P(x|z)P(y|z)$$

Flujo de inferencia
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \overset{?}{=} P(r \mid a)$	$P(r \mid t) \overset{?}{=} P(r \mid t, a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \overset{?}{=} P(l \mid e)$	$P(l \mid a) \overset{?}{=} P(l \mid e, a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \overset{?}{=} P(t \mid e)$	$P(t \mid a) \overset{?}{=} P(t \mid e, a)$ $P(t \mid l) \overset{?}{=} P(t \mid e, l)$

Flujo de inferencia
Estructuras básicas

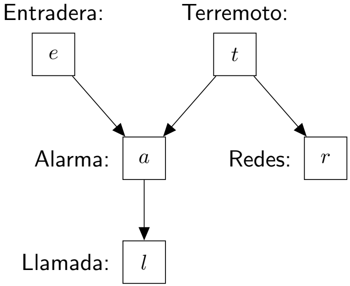


		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) \stackrel{?}{=} P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \stackrel{?}{=} P(l\|e)$	$P(l\|a) \stackrel{?}{=} P(l\|e,a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \stackrel{?}{=} P(t\|e)$	$P(t\|a) \stackrel{?}{=} P(t\|e,a)$ $P(t\|l) \stackrel{?}{=} P(t\|e,l)$

$$A \not\perp R|\emptyset$$

Flujo de inferencia

Estructuras básicas

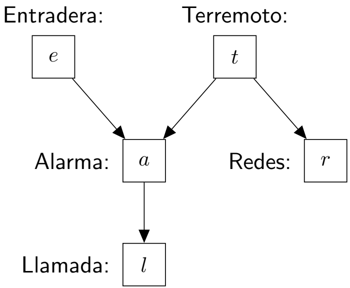


		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) = P(r\|t, a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \stackrel{?}{=} P(l\|e)$	$P(l\|a) \stackrel{?}{=} P(l\|e, a)$
Collider:	$\begin{array}{ccc} e \rightarrow & a & \leftarrow t \\ & \downarrow & \\ & l & \end{array}$	$P(t) \stackrel{?}{=} P(t\|e)$	$P(t\|a) \stackrel{?}{=} P(t\|e, a)$ $P(t\|l) \stackrel{?}{=} P(t\|e, l)$

$$A \not\perp\!\!\!\perp R|\emptyset \qquad\qquad A \perp\!\!\!\perp R|T$$

Flujo de inferencia

Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) = P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \stackrel{?}{=} P(l\|e)$	$P(l\|a) \stackrel{?}{=} P(l\|e,a)$
Collider:	$e \rightarrow a \leftarrow t$, $a \rightarrow l$	$P(t) \stackrel{?}{=} P(t\|e)$	$P(t\|a) \stackrel{?}{=} P(t\|e,a)$ $P(t\|l) \stackrel{?}{=} P(t\|e,l)$

Flujo de inferencia

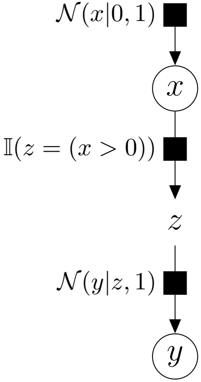
Pipe

$$x \longrightarrow z \longrightarrow y$$

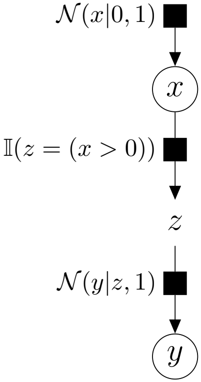
Flujo de inferencia

Pipe

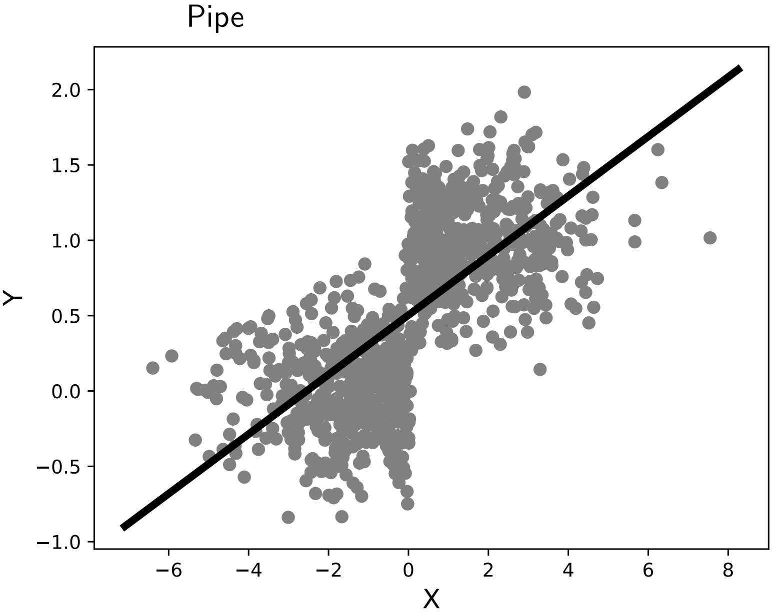
$$x \longrightarrow z \longrightarrow y$$



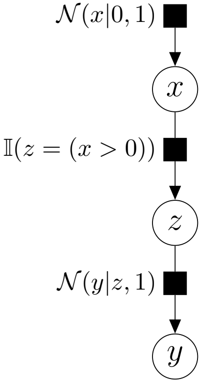
Flujo de inferencia



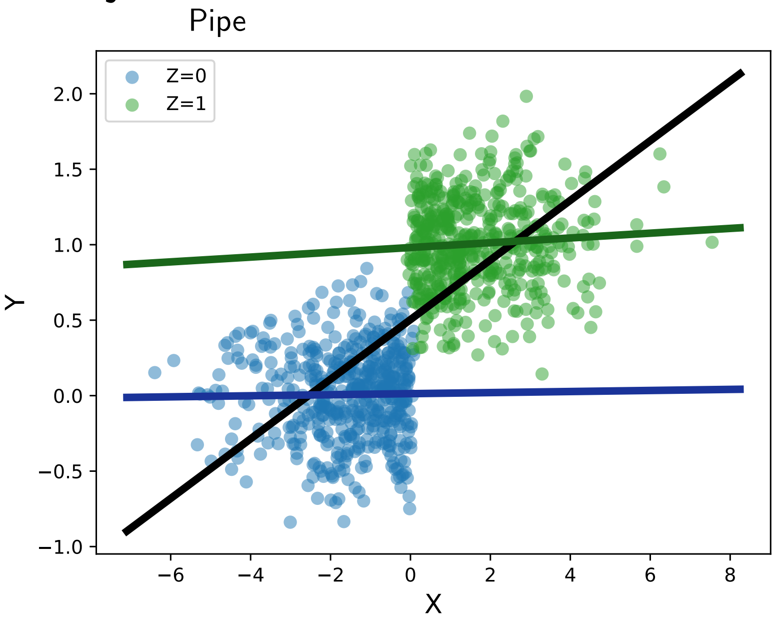
$$X \not\perp\!\!\!\perp Y | \emptyset$$



Flujo de inferencia

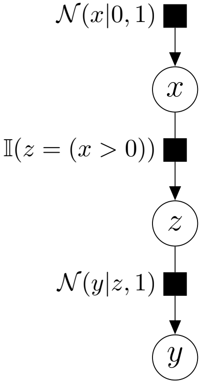


$$X \perp\!\!\!\perp Y | Z$$



Flujo de inferencia

Pipe

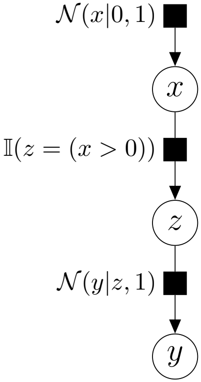


$$P(x,y|z) =$$

$$X \perp\!\!\!\perp Y | Z$$

Flujo de inferencia

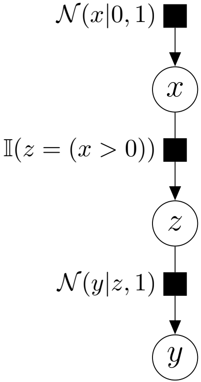
Pipe



$$P(x,y|z) = \frac{P(x,y,z)}{P(z)}$$

$$X \perp\!\!\!\perp Y | Z$$

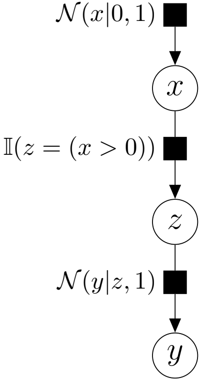
Flujo de inferencia
Pipe



$$P(x,y|z) = \frac{P(x)P(z|x)P(y|z)}{P(z)}$$

$$X \perp\!\!\!\perp Y | Z$$

Flujo de inferencia
Pipe

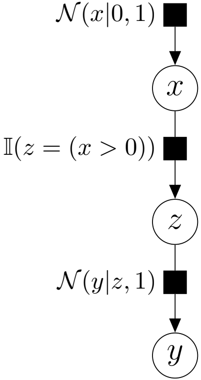


$$P(x,y|z) = \frac{P(x)P(z|x)P(y|z)}{P(z)} = \frac{P(x)P(z|x)}{P(z)}P(y|z)$$

$$X \perp\!\!\!\perp Y|Z$$

Flujo de inferencia

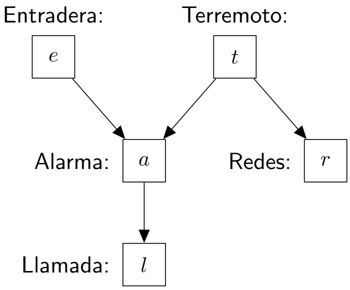
Pipe



$$P(x,y|z) = \frac{P(x)P(z|x)P(y|z)}{P(z)} = \frac{P(x)P(z|x)}{P(z)}P(y|z) = P(x|z)P(y|z)$$

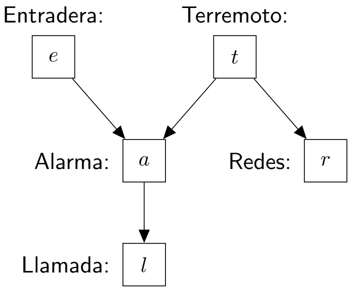
$$X \perp\!\!\!\perp Y | Z$$

Flujo de inferencia
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) = P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \stackrel{?}{=} P(l\|e)$	$P(l\|a) \stackrel{?}{=} P(l\|e,a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \stackrel{?}{=} P(t\|e)$	$P(t\|a) \stackrel{?}{=} P(t\|e,a)$ $P(t\|l) \stackrel{?}{=} P(t\|e,l)$

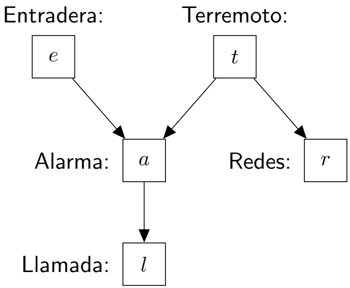
Flujo de inferencia
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) = P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \neq P(l\|e)$	$P(l\|a) \stackrel{?}{=} P(l\|e,a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \stackrel{?}{=} P(t\|e)$	$P(t\|a) \stackrel{?}{=} P(t\|e,a)$ $P(t\|l) \stackrel{?}{=} P(t\|e,l)$

$$E \not\perp\!\!\!\perp L | \emptyset$$

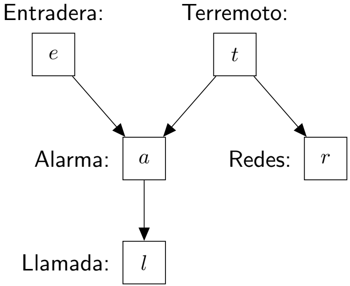
Flujo de inferencia
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r \mid a)$	$P(r \mid t) = P(r \mid t, a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \neq P(l \mid e)$	$P(l \mid a) = P(l \mid e, a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \stackrel{?}{=} P(t \mid e)$	$P(t \mid a) \stackrel{?}{=} P(t \mid e, a)$ $P(t \mid l) \stackrel{?}{=} P(t \mid e, l)$

$$E \not\perp\!\!\!\perp L|\emptyset \qquad\qquad E \perp\!\!\!\perp L|A$$

Flujo de inferencia
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r \mid a)$	$P(r \mid t) = P(r \mid t, a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \neq P(l \mid e)$	$P(l \mid a) = P(l \mid e, a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \stackrel{?}{=} P(t \mid e)$	$P(t \mid a) \stackrel{?}{=} P(t \mid e, a)$ $P(t \mid l) \stackrel{?}{=} P(t \mid e, l)$

Flujo de inferencia

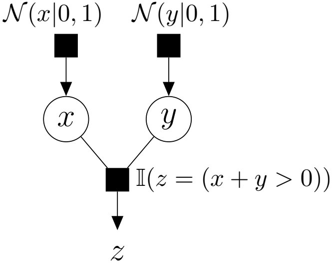
Collider

$$x \longrightarrow z \longleftarrow y$$

Flujo de inferencia

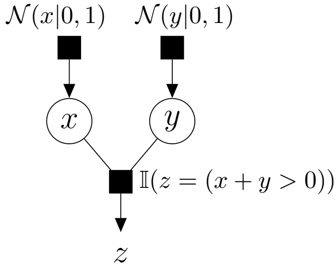
Collider

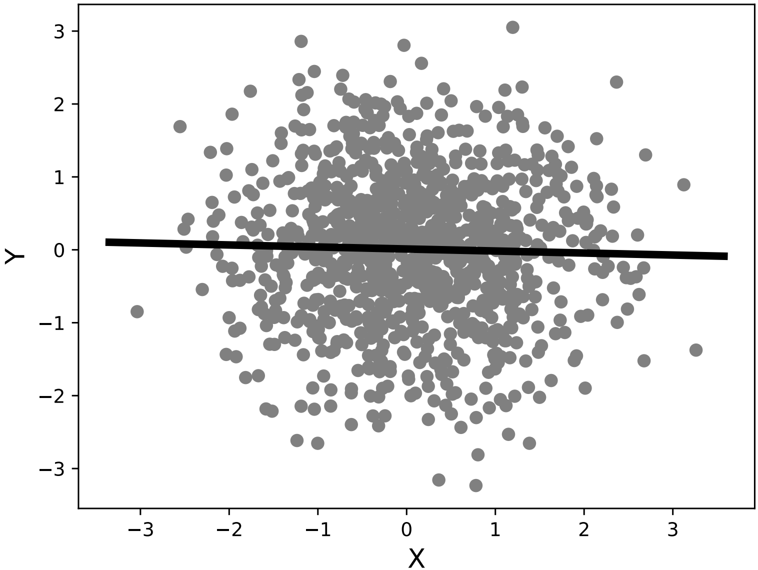
$$x \longrightarrow z \longleftarrow y$$



Flujo de inferencia

Collider

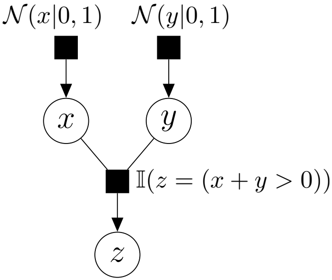


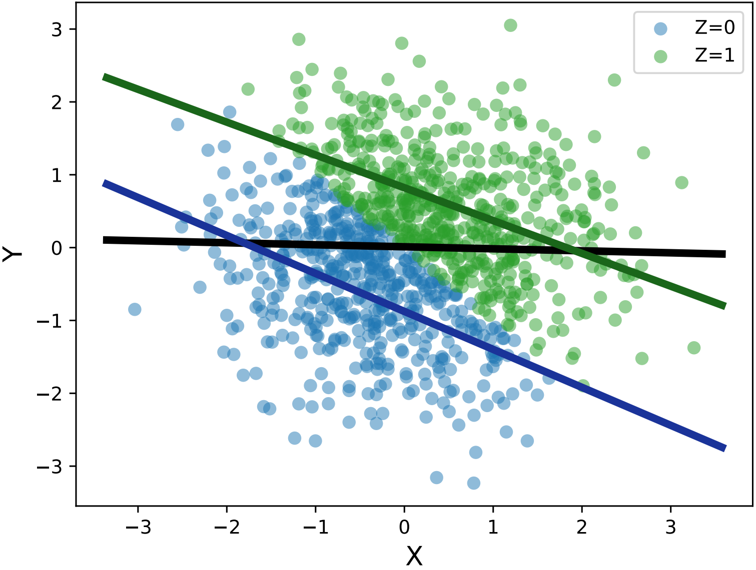


$$X \perp\!\!\!\perp Y | \emptyset$$

Flujo de inferencia

Collider

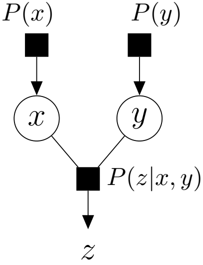




$$X \not\perp\!\!\!\perp Y | Z$$

Flujo de inferencia

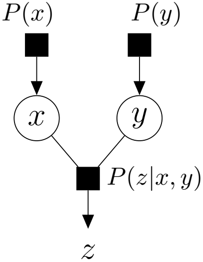
Collider



$$P(x,y) =$$

$$X \perp\!\!\!\perp Y | \emptyset$$

Flujo de inferencia
Collider

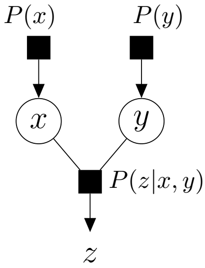


$$P(x,y) = \sum_{z} P(x,y,z)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

Flujo de inferencia

Collider

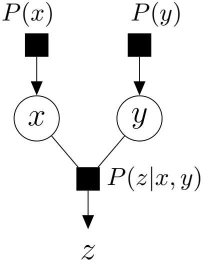


$$P(x,y) = \sum_z P(x)P(y)P(z|x,y)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

Flujo de inferencia

Collider

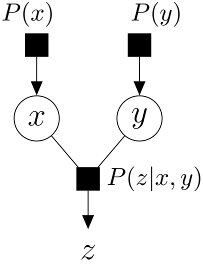


$$P(x,y) = \sum_z P(x)P(y)P(z|x,y) = P(x)P(y)\sum_z P(z|x,y)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

Flujo de inferencia

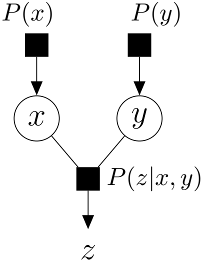
Collider



$$P(x,y) = \sum_z P(x)P(y)P(z|x,y) = P(x)P(y)\cancel{\sum_z P(z|x,y)}$$

$$X \perp\!\!\!\perp Y | \emptyset$$

Flujo de inferencia
Collider

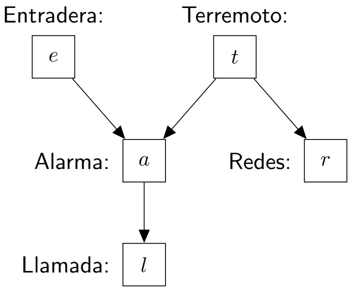


$$P(x,y) = \sum_z P(x)P(y)P(z|x,y) = P(x)P(y)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

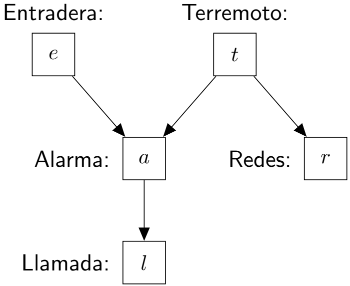
Flujo de inferencia

Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\mid a)$	$P(r\mid t) = P(r\mid t, a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \neq P(l\mid e)$	$P(l\mid a) = P(l\mid e, a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) \stackrel{?}{=} P(t\mid e)$	$P(t\mid a) \stackrel{?}{=} P(t\mid e, a)$ $P(t\mid l) \stackrel{?}{=} P(t\mid e, l)$

Flujo de inferencia
Estructuras básicas

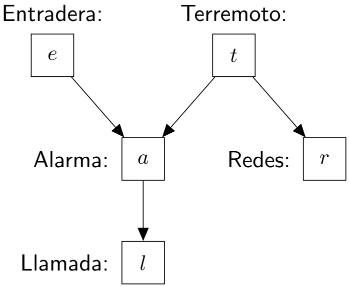


		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) = P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \neq P(l\|e)$	$P(l\|a) = P(l\|e,a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) = P(t\|e)$	$P(t\|a) \stackrel{?}{=} P(t\|e,a)$ $P(t\|l) \stackrel{?}{=} P(t\|e,l)$

$$T \perp\!\!\!\perp E | \emptyset$$

Flujo de inferencia

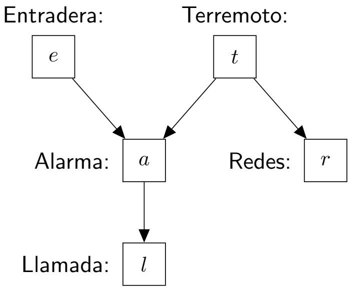
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) = P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \neq P(l\|e)$	$P(l\|a) = P(l\|e,a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) = P(t\|e)$	$P(t\|a) \neq P(t\|e,a)$ $P(t\|l) \stackrel{?}{=} P(t\|e,l)$

$$T \perp\!\!\!\perp E|\emptyset \qquad\qquad T \not\!\perp\!\!\!\perp E|A$$

Flujo de inferencia
Estructuras básicas



		Intermedio no observable	Intermedio observable
Fork:	$a \leftarrow t \rightarrow r$	$P(r) \neq P(r\|a)$	$P(r\|t) = P(r\|t,a)$
Pipe:	$e \rightarrow a \rightarrow l$	$P(l) \neq P(l\|e)$	$P(l\|a) = P(l\|e,a)$
Collider:	$\begin{array}{c} e \rightarrow a \leftarrow t \\ \downarrow \\ l \end{array}$	$P(t) = P(t\|e)$	$P(t\|a) \neq P(t\|e,a)$ $P(t\|l) \neq P(t\|e,l)$

$$T \perp\!\!\!\perp E | \emptyset \qquad\qquad T \not\!\perp\!\!\!\perp E | L$$

Flujo de inferencia

Estructuras complejas (caminos)

Hay flujo de inferencia entre los extremos de una cadena si:

(camino *d-conectado*)

- Todas las consecuencias comunes (o sus descendientes) son observables
- Ninguna otra variable es observable

Flujo de inferencia
Estructuras complejas (caminos)

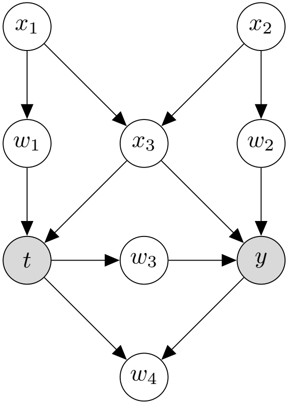
Hay flujo de inferencia entre los extremos de una cadena si:
(camino *d-conectado*)

- Todas las consecuencias comunes (o sus descendientes) son observables
- Ninguna otra variable es observable

Se cierra el flujo si está $\underbrace{\text{no } \textit{d-conectado}}_{\text{d-separado}}$

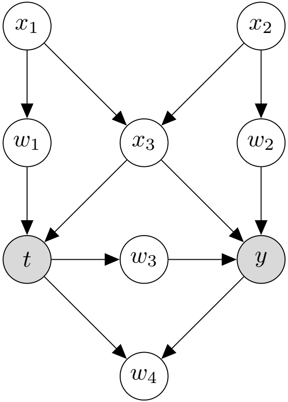
Flujo de inferencia

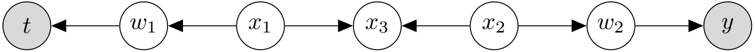
Estructuras complejas (caminos)



Flujo de inferencia

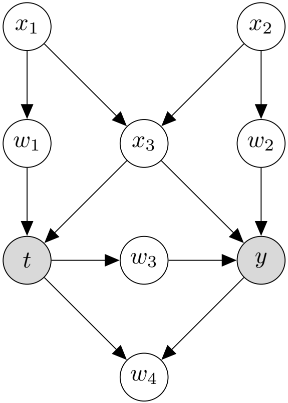
Estructuras complejas (caminos)

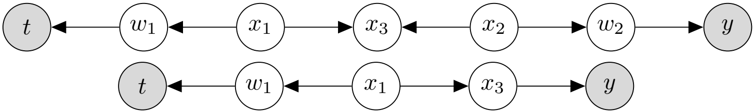




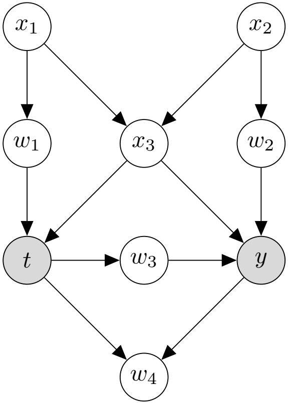
Flujo de inferencia

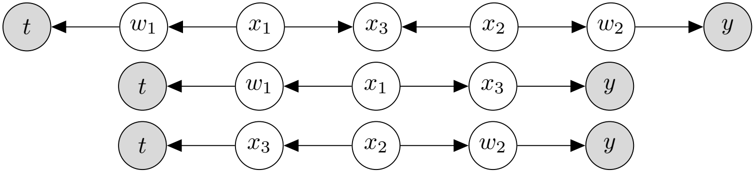
Estructuras complejas (caminos)





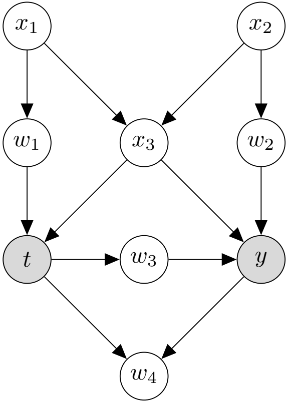
Flujo de inferencia
Estructuras complejas (caminos)

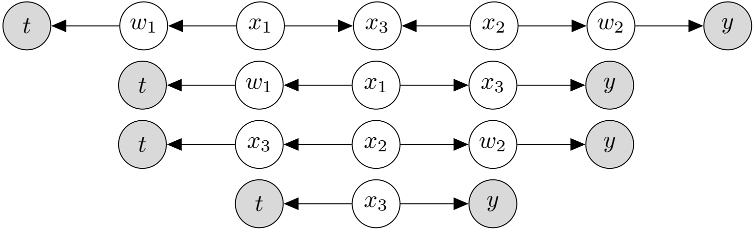




Flujo de inferencia

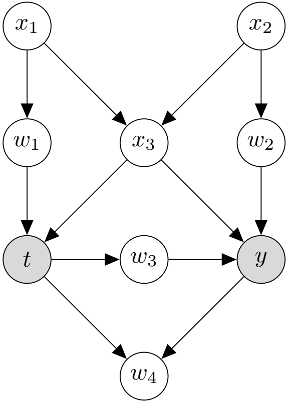
Estructuras complejas (caminos)

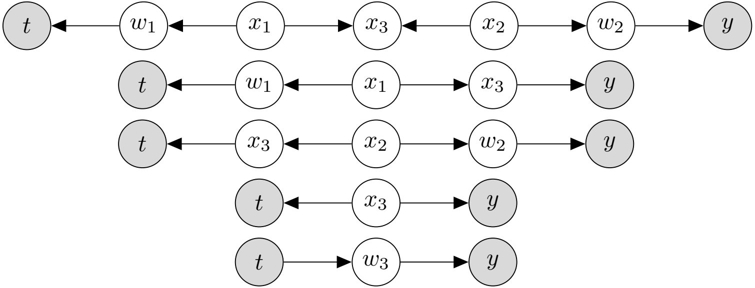




Flujo de inferencia

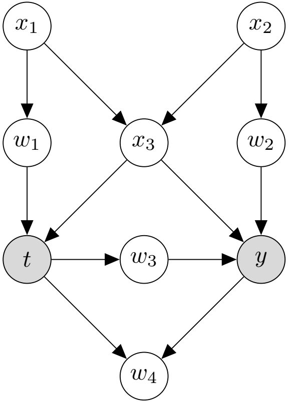
Estructuras complejas (caminos)

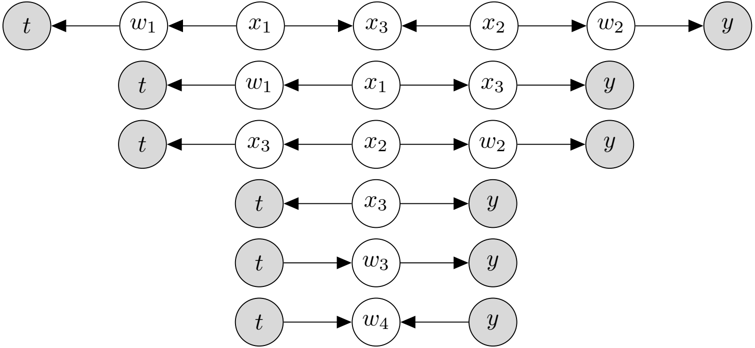




Flujo de inferencia

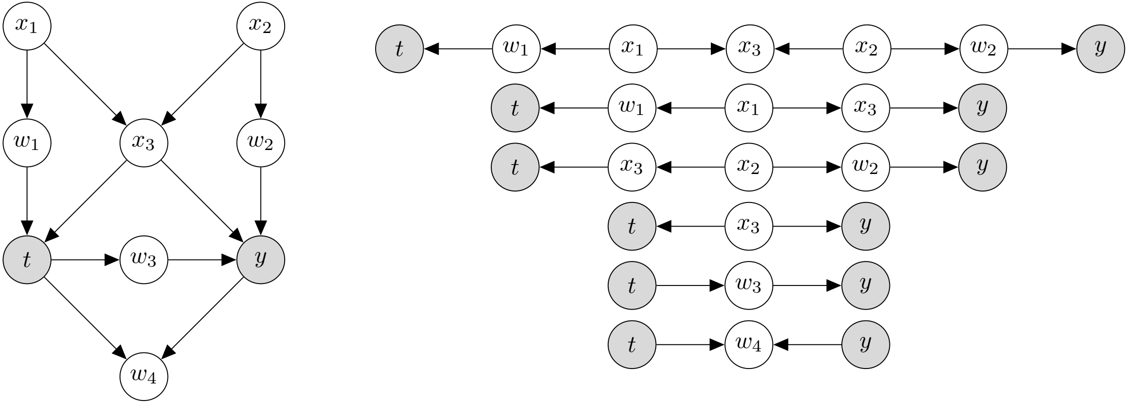
Estructuras complejas (caminos)





Flujo de inferencia

Estructuras complejas (caminos)



Para hacer predicciones "causales" necesitamos
eliminar el flujo de inferencia que circula por los caminos no causales

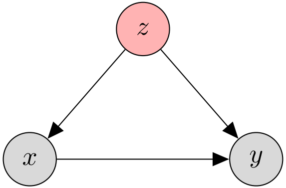
Flujo de inferencia
Variables de control

Backdoor criterion

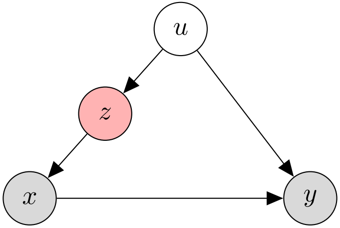
Dado un conjunto de variable $\boldsymbol{Q}$ tal que:

1. $\boldsymbol{Q}$ cierra todos los caminos traseros de T a Y
2. $\boldsymbol{Q}$ no contiene ningún descendiente de T

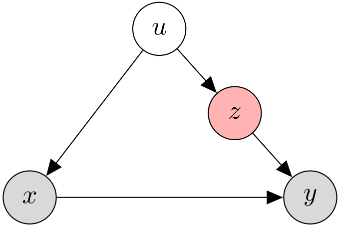
Controles buenos



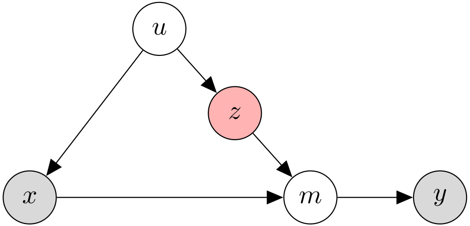
Controles buenos



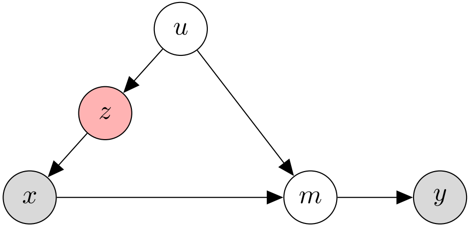
Controles buenos



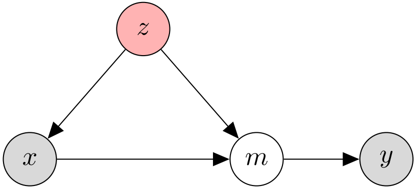
Controles buenos



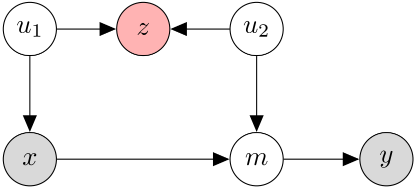
Controles buenos



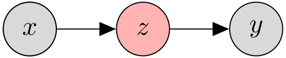
Controles buenos



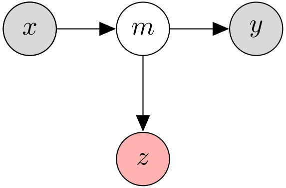
Controles malos



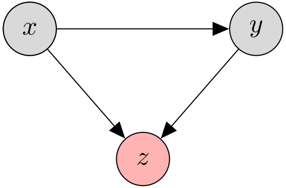
Controles malos



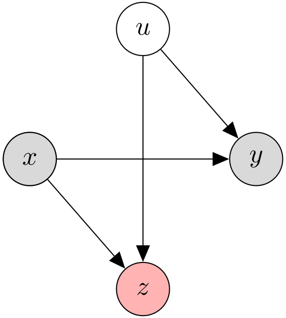
Controles malos



Controles malos

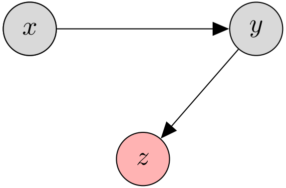


Controles malos



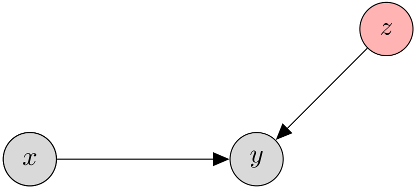
Controles malos

Sesgo de selección



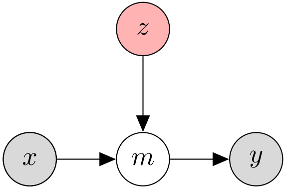
Controles neutrales

Mejoran precisión



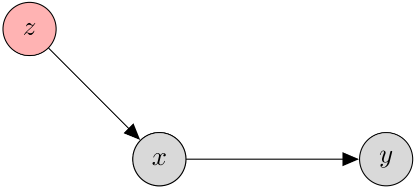
Controles neutrales

Mejoran precisión

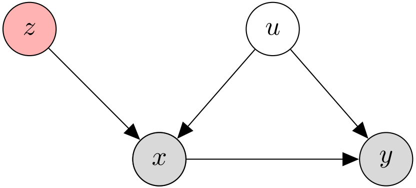


Controles neutrales

Reducen precisión



Controles malos



$p = b$

Laboratorios de
Métodos Bayesianos